



Bahir Dar Institute of Technology

Faculty of Electrical and Computer Engineering

CHAPTER ONE

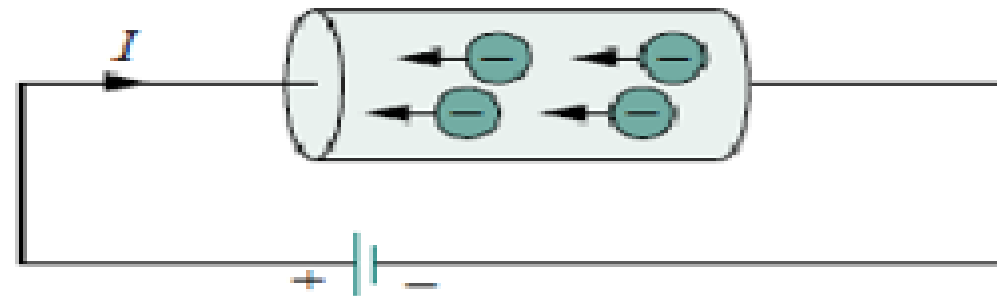
BASICS OF ELECTRICITY

Charge, Current and Voltage

Electric Charges

- We know from elementary physics that all matter is made of fundamental building blocks known as atoms and that each atom consists of **electrons, protons, and neutrons**.
- Charge is a property of matter, just like mass, volume, or density. It is measurable.
- We also know that the charge **e** on an electron is negative while a proton carries a positive charge and equal in magnitude to **1.602×10^{-19} Coulombs**
- When a conducting wire (consisting of several atoms) is connected to a battery (a source of electromotive force), the charges are compelled to move; positive charges move in one direction while negative charges move in the opposite direction. This motion of charges creates electric current.

- It is conventional to take the current flow as the movement of positive charges, that is, opposite to the flow of negative charges, as Fig. below illustrates.



- One coulomb is the total charge possessed by $6.25 \times 10^{18} e^{-}$
 $1C = 6.24 \times 10^{18}$ electrons
- The total charge Q , expressed in coulombs, for a given number of electrons is stated in the following formula:

$$Q = \frac{\text{number of electrons}}{6.24 \times 10^{18} \text{ electrons} / C}$$

Examples

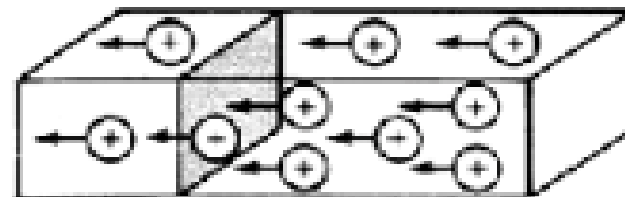
- 1 How many coulombs of charge do 93.8×10^{16} electrons represent?
- 2 How many electrons does it take to have 3 C of charge?
- 3 After 10.61×10^{13} electrons are added to a metal plate, it has a negative charge of 3C. What was its initial charge in coulombs?

Answer

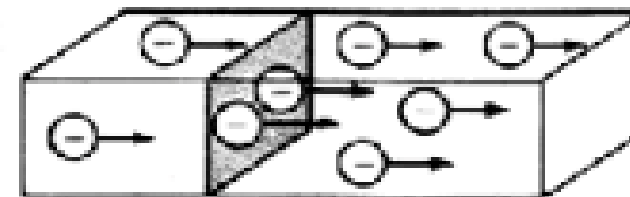
- 1 0.15C
- 2 18.75×10^{18} electrons
- 3 $+14 \mu\text{C}$

CURRENT

- Electric current $i = dq/dt$. The unit of ampere can be derived as $1 \text{ A} = 1\text{C/s}$.
- A direct current (dc) is a current that remains constant with time.
- An alternating current (ac) is a current that varies sinusoidally with time. (reverse direction)



(a)
Positive ions



(b)
Negative ions

- The more electrons per second that pass through the circuit, the greater is the current.
- Thus, current is the rate of flow (or rate of movement) of charge and the symbol for current is I .
Expressed mathematically,

$$I = \frac{Q}{t}$$

where Q is the charge and t is the time interval over which it is measured. Through algebraic manipulations, we have

$$Q = It \text{ [coulombs]} \text{ and } t = \frac{Q}{I} \text{ [seconds, S]}$$

- The SI unit of current is ampere(A), and
- From this, we get that , One ampere (1 A) is the amount of current that exists when a number of electrons having a total charge of one coulomb (1 C) move through a given cross-sectional area in one second (1 s).

Example 1. A conductor has a constant current of 5 A. How many electrons pass a fixed point on the conductor in one minute?

Total no. of charges pass in 1 min is given by

$A = (5 \text{ C/s})(60 \text{ s/min}) = 300 \text{ C/min}$ Total no. of electronics pass in 1 min is given

$$\frac{300 \text{ C / min}}{1.602 \times 10^{-19} \text{ C / electron}} = 1.87 \times 10^{21} \text{ electron/min}$$

Exercise

- 1 If 840 coulombs of charge pass through the give point during a time interval of 2 minutes, what is the current?
- 2 If 12.48×10^{20} electrons pass a certain point in a circuit in 2.5 s, what is the current in amperes?
- 3 If there are 8 A of current through the filament of a lamp, how many electrons are move through the filament in 1.5 s?

Voltage

- To move the electron in a conductor in a particular direction requires some work or energy transfer.
- This work is performed by an external electromotive force (emf), typically represented by the battery
- This emf is also known as voltage or potential difference.
- Voltage (or potential difference) is the energy required to move a unit charge through an element, measured in volts (V).
- Voltage is, therefore, an indication of how much energy is involved in moving a charge between two points in an electrical system.

- By definition, the voltage between two points is one volt if it requires one joule of energy to move one coulomb of charge from one point to the other. In equation form.

$$V = \frac{W}{Q} [\text{Volts}, V]$$

where W is energy in joules, Q is charge in coulombs, and V is the resulting voltage in volts.

- Through algebraic manipulations, we have

$$W = QV [\text{Joules}, J] \text{ and } Q = \frac{W}{V} [\text{coulombs}, C]$$

- Note carefully that voltage is defined between points.
- For the case of the battery, for example, voltage appears between its terminals.
- Thus, voltage does not exist at a point by itself; it is always determined with respect to some other point. _ _ _ _

Example

- 1 If it takes 35 J of energy to move a charge of 5 C from one point to another, what is the voltage between the two points?
- 2 The voltage between two points is 19 V. How much energy is required to move 67×10^{18} electrons from one point to the other?

Answer

1) 7V

2) 204J

RESISTANCE (R)

- The flow of charge through any material encounters an opposing force similar in many respects to mechanical friction.
- This opposition, due to the collisions between electrons and other atoms in the material, which converts electrical energy into another form of energy such as heat, is called the **resistance (R)** of the material.
- It may be defined as the property of a substance due to which it opposes (or restricts) the flow of electricity (i.e., electrons) through it.
- Resistance is expressed in ohms, symbolized by the Greek letter omega (Ω). The circuit symbol for resistance is



- The resistance of any material with uniform cross-sectional area depends on the following factors:
 - Type of material, Length of material, Cross-sectional area and Temperature

- The factors governing the resistance of a conductor at a fixed temperature of 20°C (room temperature) may be summarized mathematically as follows:

$$R = \frac{\rho l}{A} \text{ [Ohms, } \Omega \text{]}$$



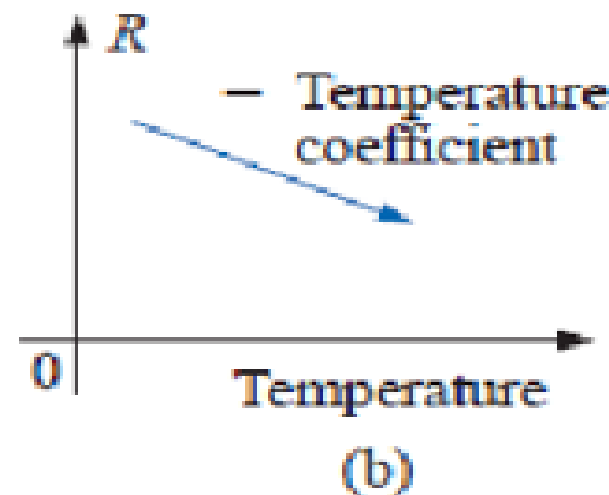
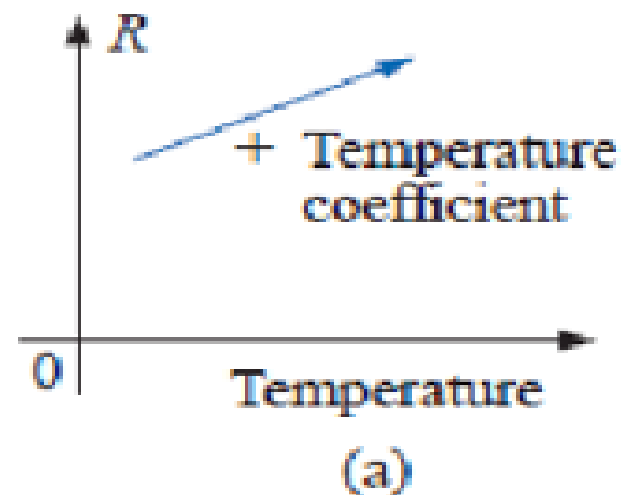
where:

- ρ = the resistivity of the material in ohm metres, (Ω -m)
- l = the length of the material, in m
- A = the cross-sectional area of the material, in sq. meter.

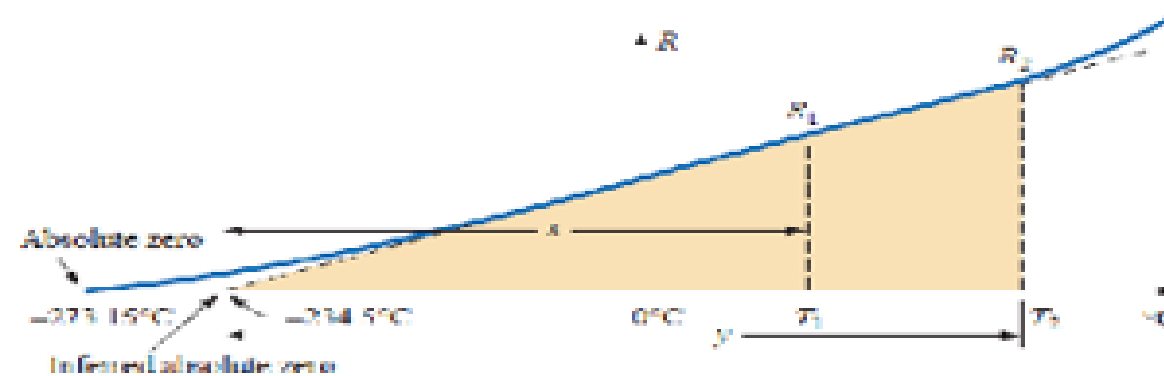
- In the above equation the lowercase Greek letter rho (ρ) is the constant of proportionality and is called the resistivity of the material.

TEMPERATURE EFFECTS

- Temperature has a significant effect on the resistance of conductors, semiconductors, and insulators.
- for good conductors, an increase in temperature will result in an increase in the resistance level. Consequently, conductors have a positive temperature coefficient. for semiconductor materials, an increase in temperature will result in a decrease in the resistance level. Consequently, semiconductors have negative temperature coefficients.



- As with semiconductors, an increase in temperature will result in a decrease in the resistance of an insulator. The result is a negative temperature coefficient.
- Figure below reveals that for copper (and most other metallic conductors), the resistance increases almost linearly (in a straight-line relationship) with an increase in temperature.



- An equation for determining the resistance at any temperature within operating limits can be obtained by approximating the curve of the above Fig. by the straight dashed line that intersects the temperature scale at -234.5°C .

- Let x equal the distance from $-234.5\text{ }^{\circ}\text{C.}$ to T_1 and y the distance from $-234.5\text{ }^{\circ}\text{C.}$ to T_2 , as shown in

$$\frac{x}{R_1} = \frac{y}{R_2}$$

$$\frac{234.5 + T_1}{R_1} = \frac{234.5 + T_2}{R_2}$$

- For different conducting materials, the intersection of the straight-line approximation will occur at different temperatures.

Inferred absolute temperatures (T_i).

Material	$^{\circ}\text{C}$
Silver	243
Copper	-234.5
Gold	-274
Aluminum	236
Tungsten	204
Nickel	147
Iron	162
Nichrome	2,250
Constantan	-125,000

Temperature Coefficient of Resistance (Reading Assignment)

Example

If the resistance of a copper wire is $50\ \Omega$ at $20\ ^\circ\text{C}$ what is its resistance at $100\ ^\circ\text{C}$ (boiling point of water)?

$$\frac{R_1}{234.5\ ^\circ\text{C} + 20\ ^\circ\text{C}} = \frac{R_2}{234.5\ ^\circ\text{C} + 100\ ^\circ\text{C}}$$
$$R_2 = \frac{(50\ \Omega)(334.5\ ^\circ\text{C})}{254.5\ ^\circ\text{C}} = 65.72\ \Omega$$

Power and Energy

- Energy is the ability to do work, and Power is the rate at which energy is used.
- Power (P) is a certain amount of energy (W) used in a certain length of time (t), expressed as follows:

$$P = \frac{W}{t} [Watt]$$

- The SI unit of power is the watt(W) and that of Energy is joules(J)
- N.B:- you can see that one watt equals one joule per second.
- In electrical power, we need expressions for P in terms of electrical quantities.
- We know that, voltage is defined as work per unit charge and current as the rate of transfer of charge,

- From the above Equation, $W = QV$ and Substituting this into Equation($P = W/t$) yields $P = (QV)/t = V(Q/t)$. Replacing Q/t with I , we get

$$P = VI$$

- Additional relationships are obtained by direct substituting of ohms law ($V= IR$ and $I=V/R$) into Equation 2.7:

$$P = I^2 R$$

$$P = \frac{V^2}{R}$$

where: P = is Power delivered to the load.
 V = is the voltage across the load
 I = is the current through the load
 R = is the resistance value of the load.

Coulomb's Law and Electric Field

Coulomb's Law

- It has been determined by experimentation that unlike charges attract, and like charges repel.
- The force of attraction or repulsion between two charged bodies Q_1 and Q_2 can be determined by Coulombs law: Mathematically,

$$F = k \frac{Q_1 Q_2}{r^2}$$

where Q_1 and Q_2 are the charges in coulombs, r is the distance between them in meters, and k is constant, $k = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$.

- Coulombs law applies to aggregates of charges as well as to individual electrons within the atom.

- In particular, note the squared r term in the denominator, resulting in rapidly decreasing levels of F for increasing values of r .
- Because of this relationship, electrons in outer orbits are less strongly attracted to the nucleus than those in inner orbits; i.e. they are less tightly bound to the nucleus than those close by.
- Valence electrons are the least tightly bound and will, if they acquire sufficient energy, escape from their parent atoms.
- The amount of energy required to escape depends on the number of electrons in the valence shell.
- If an atom has only a few valence electrons, only a small amount of additional energy is needed
- For example, for a metal like copper, valence electrons can gain sufficient energy from heat alone (thermal energy), even at room temperature, to escape from their parent atoms and wander from atom to atom throughout the material .

Example :

1) Positive charges $Q_1 = 2\mu\text{C}$ and $Q_2 = 12\mu\text{C}$ are separated center to center by 10 mm. Compute the force between them. Is it attractive or repulsive?

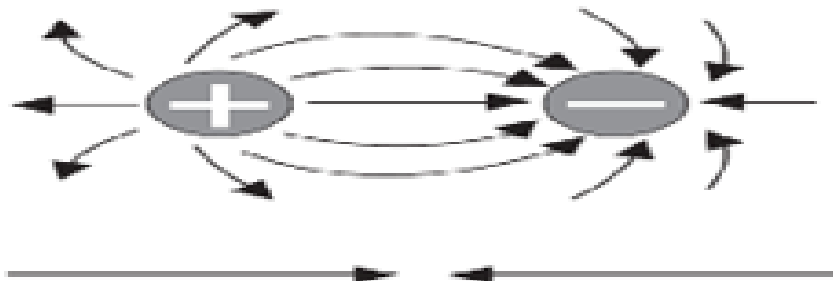
Answers:

1) 2160 N, repulsive;

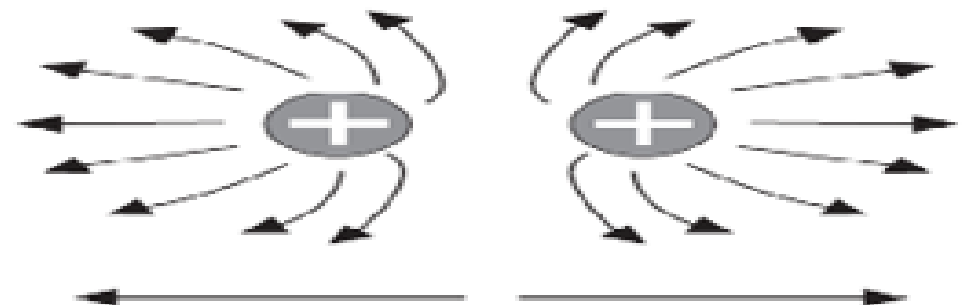
Electric Fields

- Around a charged body, there will be invisible lines of force represent an invisible electrical field that causes the attraction and repulsion.
- These invisible lines of force around the charged body is known as an Electric Field.
- The electric field around a charged body is represented by electric flux lines.
- The denser the flux lines, the stronger the electric field.

Unlike Charges Attract



Like Charges Repel



- By definition, the electric field strength at a point is the force acting on a unit positive charge at that point. i.e. $E = \frac{F}{Q}$ where E is electric field, Q is a unit charge and F is the exerted force on unit charge Q .

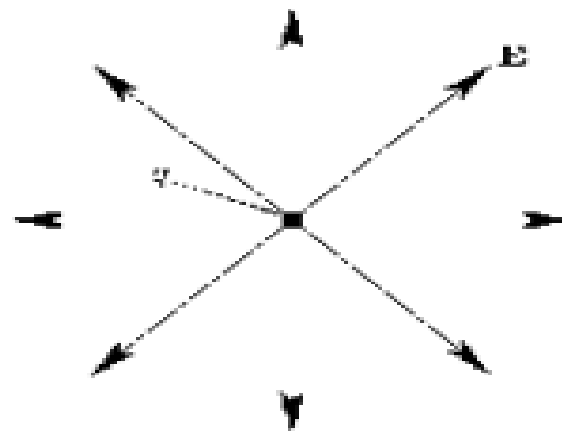
- The force exerted on a unit positive charge ($Q_2 = 1C$) by a charge Q_1 , r meter away is given by, using coulomb's law:

$$E = \frac{F}{Q_2} = k \frac{Q_1 Q_2}{r^2} / Q_2, \quad E = k \frac{Q_1}{r^2}$$

This shows that:

- $E \propto Q_1$ - the bigger the charge, the stronger field.
- $E \propto \frac{1}{r^2}$ - another inverse square relationship. The further you are from the charge, the weaker the field strength.
- Hence, E at any point from Q is directly proportional to the magnitude of the charge and inversely proportional to the distance squared from the charge.

- Thus, field-lines determine the magnitude, as well as the direction, of the electric field. In particular, the field is strong at points where the field-lines are closely spaced, and weak at points where they are far apart.



- **N.B:-** Electric flux lines always extends from positive charged body to a negative or terminate perpendicular to the charged surface, and never intersect.

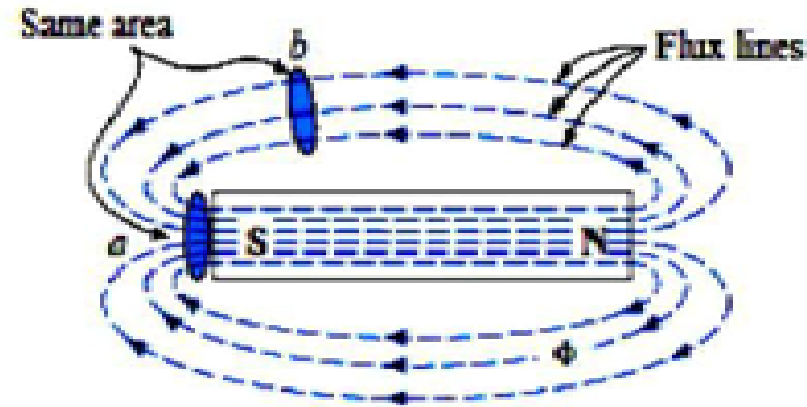
Example:- Find the electric field strength 50cm from positive charge of $Q = 1 \times 10^{-4} \text{ C}$, ($k = 9 \times 10^9 \text{ Nm}^2 / \text{C}^2$)

Magnetic Field and Faraday's Law

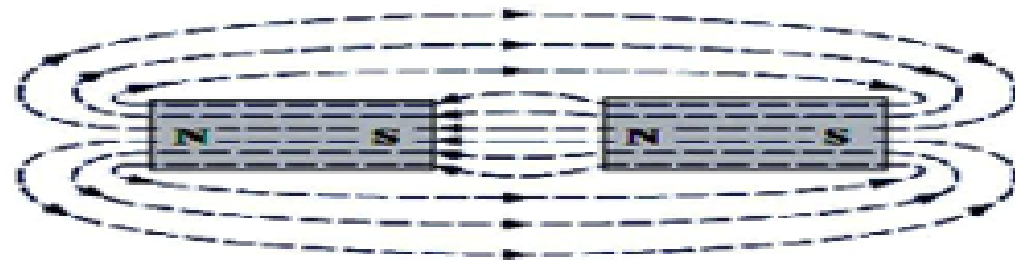
Magnetic Field

- In the region surrounding a permanent magnet there exists a magnetic field, which can be represented by magnetic flux lines similar to electric flux lines.
- Magnetic flux lines, however, do not have origins or terminating points as do electric flux lines but exist in continuous loops.
- It radiates from the north pole to the south pole, returning to the north pole through the metallic bar. as shown below. 2
- The strength of a magnetic field in a particular region is directly related to the density of flux lines in that region.

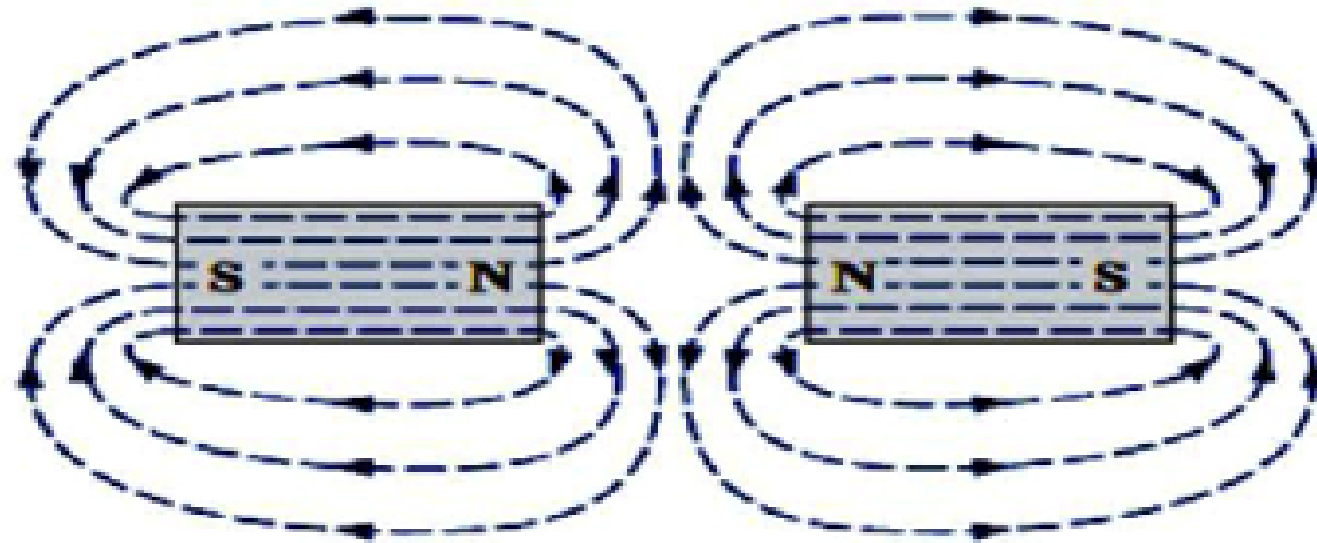
- For example, the magnetic field strength at a is twice that at b.



- If unlike poles of two permanent magnets are brought together, the magnets will attract, and the flux distribution will be as shown below.

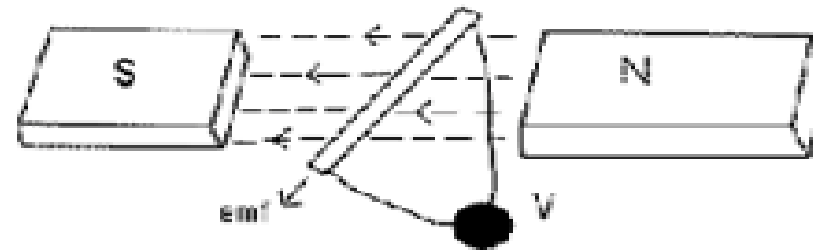


- If like poles are brought together, the magnets will repel, and the flux distribution will be as shown .



Faraday's Law's of Electromagnetic Induction

- Faraday's first law state that whenever a conductor are placed in a varying magnetic field emf are induced which is called induced emf.or whenever a conductor is rotate in a magnetic field emf are induce which are induced emf.



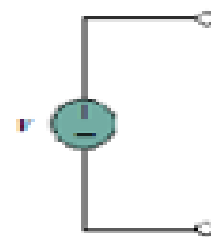
- Faraday's Second law state that the emf induced in a circuit is directly proportional to the time rate of change of the magnetic flux through the circuit.

$$E = N \frac{d\phi}{dt}$$

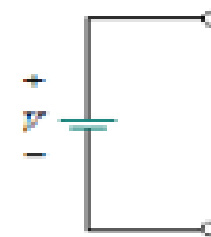
where N represents the number of turns of the coil and $\frac{d\phi}{dt}$ is the instantaneous change in flux (in webers).

Electric Sources (Ideal/Real; Independent/Dependent)

- There are two kinds of sources: independent and dependent sources.
- An ideal independent source is an active element that provides a specified voltage or current that is completely independent of other circuit variables.
- Physical sources such as batteries and generators may be regarded as approximations to ideal voltage sources.
- Figure shows below the symbols for independent voltage sources. (a) used for constant or time-varying voltage, (b) used for constant voltage (dc).

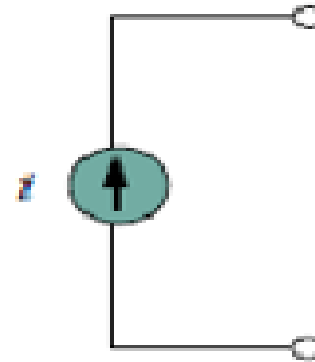


(a)



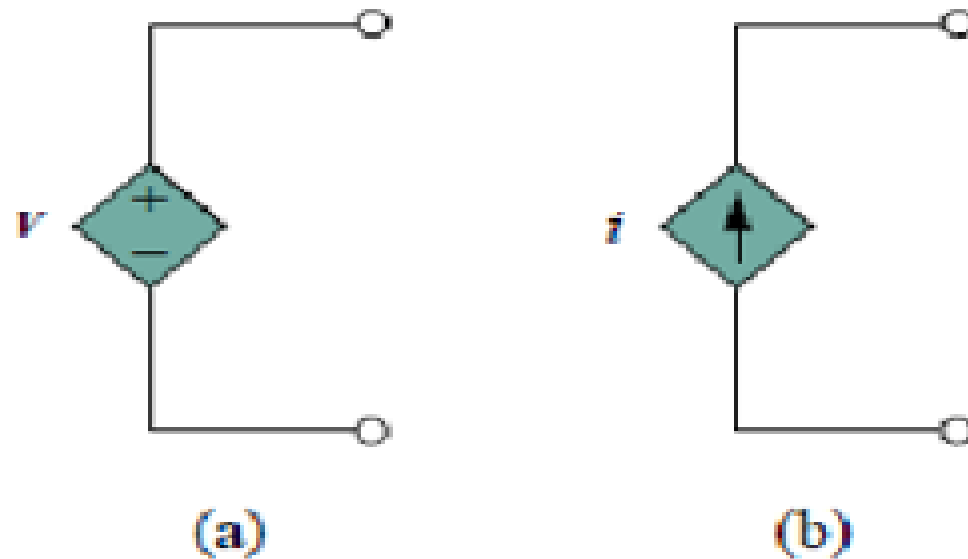
(b)

- Similarly, an ideal independent current source is an active element that provides a specified current completely independent of the voltage across the source.



- An ideal dependent (or controlled) source is an active element in which the source quantity is controlled by another voltage or current.
- Dependent sources are useful in modeling elements such as transistors, operational amplifiers and integrated circuits.

- Dependent sources are usually designated by diamond-shaped symbols, as shown in Fig. below.
- Symbols for: (a) dependent voltage source, (b) dependent current source.



THANK YOU!!!